
CASE STUDY OF COINCHECK HACK INCIDENT (2018)

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EXECUTIVE SUMMARY

On 26 January 2018, one of the most significant breaches within the history of cryptocurrencies happened at a Japanese exchange, Coincheck Inc., also a Tokyo-based company. The attackers managed to hack and succeed in stealing about 523 million NEM, or XEM, worth about ¥58 billion at the time, a total of about US\$530 million, from the online wallets of this exchange. This major incident highlights the inadequacies in the security methodology used at this exchange regarding the storage of its digital assets.

The incident resulted from poor wallet management practices, specifically the practice of holding a considerable amount of funds in a hot wallet that requires only a signature approval in a multi-signature wallet system or a hardware-based key management approach to protect the keys from unauthorized access. This approach ultimately led to a vulnerable system, which allowed all funds to be transferred once they gained entry. It appears from publicly available investigations that they might have gained entry into the system through a phishing attack on the internal employees of the company in question; however, they did not share the system logs or the malware.

Also, it had adverse financial implications for more than 260,000 customers who had their assets frozen immediately after detection of being compromised. Even though there was no conclusive public data available that supported customer data or login being compromised, it led to freezing of withdrawal or trade functions for an extended period of time. Eventually, Coincheck repaid all customers from its corporate account, thereby reducing losses for its customers but leaving lasting damage to its reputation.

In forensic terms, it can be said that blockchain transparency made it possible to track the stolen money flow within the NEM blockchain using free blockchain explorers. As it often happens with cyber thefts, the stolen money was immediately split across many addresses to prevent tracking. Later, those funds were laundered in dark markets. Although some of those actions were detected, which led to some of those responsible being arrested, it is a matter of regret that most of the stolen money was never retrieved.

This incident led to the Coincheck crisis being addressed immediately by the Financial Services Agency in Japan, which resulted in a business improvement order. This incident also led to the acquisition of Coincheck by Monex Group and changes in the security processes used by exchanges in Japan. This case is recognized for the essential role that sound wallet design and key management solutions, real-time transaction monitoring capabilities, and readiness for responding to incidents have at cryptocurrency exchange platforms. The key takeaways from the hack that occurred on the Coincheck exchange are contributing factors for the development of current best practices for cryptocurrency exchange platforms.

Keywords: Cryptocurrency Exchange Security, Coincheck Hack, Blockchain Forensics, Hot Wallet Vulnerabilities, Incident Response

I. INTRODUCTION

Since the emergence of Bitcoin, several cryptocurrencies have been developed, launched into the market, and drawn significant attention. Cryptocurrency (Bitcoin) is a completely decentralized digital currency that was based on blockchain technology. Even though all transaction records are public, cryptocurrency payments are anonymous unless the addresses and transactions can be matched to actual identities. This online anonymizing technology has considerable potential to increase convenience and improve social welfare. However, there is a dark side to the new technology when it is combined with another online anonymizing technology called the Tor network, in which users' messages are routed through a series of relays serving as a buffer between the users and the websites that they visit. It is difficult to identify the location of a website's visitor. These online anonymizing technologies have given rise to illicit online marketplaces called crypto markets. The most popular products sold on crypto markets are illicit drugs, since crypto market administrators can hide the location of their website servers and thus avoid law enforcement agencies. For that reason, crypto markets have received a great deal of attention from the media, government authorities, and law enforcement agencies, and many studies have been conducted.

Until January 26, 2018, we saw a cybercrime using these two new technologies that had yet to be realized. 58 billion yen (\$530 million) worth of NEM, a cryptocurrency, was fraudulently accessed, and then stolen from the Coincheck Exchange, headquartered in Japan. This hacking incident is unprecedented, not only because it was one of the world's largest cryptocurrency heists, but also because the stolen NEM was sold and money laundered on a crypto market opened by the hackers themselves. The hackers offered to swap stolen NEM for other cryptocurrencies, including Bitcoin and Litecoin, at a 15% discount against the market

rate at the time of the transaction. Moreover, the hacking incident caused significant price drops and swings in NEM and other cryptocurrencies. Three years later, on January 22, 2021, the MPD of Japan announced¹ that 31 people were charged with allegedly exchanging NEM cryptocurrency for other cryptocurrencies while knowing the assets were stolen. The trading in crypto markets involving 31 people is estimated to have totaled approximately 20 billion yen (using the exchange rate at the time of the theft). Although there has been considerable effort by authorities and volunteers alike, the hackers have not been arrested yet, and the remaining NEM, upwards of 30 billion, has not been traced. Despite around one-third of the stolen NEM being traced by the policing effort, few scholarly studies have investigated this hacking incident or how the stolen NEM was money laundered.

Therefore, this study investigates how the stolen NEM was money laundered to Bitcoin on a crypto market by investigating NEM and Bitcoin transactions.

Case Study Selection Rationale

The Coincheck 2018 breach was selected due to the availability of publicly verifiable information, including official disclosures, regulatory reports, and NEM blockchain transaction records. This transparency enables reliable timeline reconstruction, impact assessment, and forensic fund tracing without reliance on non-public data. The incident also led to major regulatory and security reforms in Japan, making it a relevant case for exchange security analysis.

II. INCIDENT ANALYSIS OF COINCHECK HACK 2018

In late January 2018, a major cryptocurrency exchange, Coincheck, a company that operates in Tokyo, experienced a remarkably large exchange hack when unknown individuals used their

exchange for unauthorized transactions of around 523-534 million of their NEM (XEM) tokens, which were worth a total of ¥58 billion (~\$530-\$534) at that time, from exchange-controlled accounts to their own.



Fig:-1. This gesture indicates a **public apology or expression of remorse**, commonly seen in Japanese corporate culture following a serious incident or failure. The setting appears official, suggesting accountability, transparency, and acknowledgment of responsibility toward the public and media.

“The attack vector was functional, in that Coincheck chose to maintain a large amount of customer-held NEM in what was effectively a hot wallet, as opposed to maintaining customer-held assets in a cold/offline environment. This factor of maintaining customer-held assets in a hot wallet has been identified as the attack vector in regulatory and industry analyses.”

Technical tracing on the blockchain of the NEM network revealed that the funds were consolidated for a series of withdrawals to a group of attacker addresses, which were then attempted to be tagged and traced by the NEM community and exchanges in order to limit their further tracing and detection, but most of it was laundered nevertheless.

"Its initial operational response involved preventing customer withdrawals and trading for most of the tokens, announcing news of the hack, and paying back those who were affected; in order to pay back around 260,000 of its customers, Coincheck used its own funds, which amounted to about 46.3-46.6 billion yen."

Regulators acted quickly in this case, with orders for business improvement, submission of a plan for remediation, and driving a broader examination of exchange security and custody practices in Japan, which resulted in a tougher set of operating requirements for exchanges, followed by a purchase of Coincheck by Monex Group in April 2018.

Root cause analysis also highlighted a lack of internal control over:

- Cold storage segregation
- Multi-sig and key isolation
- Transaction monitoring and alerting
- Corporate governance related to custody
- Auditing and risk management, which allowed a breach that would otherwise have been a low-risk event to result in a high-value loss.

Forensic examiners and incident responders are reminded of three takeaways from this incident: (1) never aggregate significant customer funds in a hot wallet, and use multi-sig and cold wallet storage for major holdings; (2) embrace robust key management, least privilege access, and automated chain of custody logging as well as real-time transaction anomaly analysis; and (3) develop playbooks for customer remediation and working with law enforcement and chain analysis collaborators—since customer compensation, communication, and regulator cooperation all proved important in remediation, reputation, and survival.



Fig:-2 Coincheck Headquarters, Tokyo, During Post-Breach Media Coverage (2018)

i. TIMELINE OF THE COINCHECK

Date & Time (JST)	Event	Detailed Description
Before Jan 26, 2018	Weak Security Practices	Coincheck stored 523 million NEM tokens in a single hot wallet without multi-signature security, making it extremely vulnerable.
Jan 26, 2018 — 02:57 AM	Initial Breach (Attack Begins)	Hackers gained unauthorized access to Coincheck's network and drained the entire NEM hot wallet . Tokens were transferred to a hacker-controlled address.
11:25 AM	Breach Detected	Coincheck staff noticed a sudden drop in NEM holdings. Internal alarms were triggered.
11:58 AM	Suspension of NEM Trading	Coincheck halted all NEM deposits and withdrawals.
12:07 PM	Full Withdrawal Freeze	Trading of all cryptocurrency withdrawals other than Bitcoin was disabled to prevent further losses.
3:00 PM (Approx.)	Public Announcement	Coincheck tweeted and released a public notice confirming the incident. News quickly spread across global crypto communities.
Evening – Jan 26, 2018	Impact Assessment & Government Alert	Coincheck confirmed that 523 million XEM tokens (~\$530M) were stolen. The Japanese Financial Services Agency (FSA) was notified.
Jan 27–28, 2018	On-chain Tracking Begins	NEM Foundation worked with exchanges worldwide. A “ tracking tag ” was added to stolen NEM to identify tainted coins.
Jan 29 – Feb 2, 2018	Attacker Laundering Efforts	Stolen NEM was moved to darknet marketplaces and sold at discounts as the hackers attempted to cash out.
Feb 2, 2018	FSA Issues Business Improvement Order	Japan's regulator forced Coincheck to submit a detailed report and improve security standards.
Feb 2018	Refund Strategy Announced	Coincheck committed to compensating all affected customers using internal company funds
Mar – Apr 2018	Customer	260,000+ users were refunded at a fixed rate of

	Reimbursements Completed	¥88.549 per NEM.
Apr 6, 2018	Coincheck Acquired by Monex Group	Monex (a major Japanese brokerage) bought Coincheck for ¥3.6 billion and began rebuilding its security infrastructure.
2018–2020	Ongoing Laundering Investigations	Authorities traced transactions across multiple countries. Some funds were seized but majority remain unrecovered.
2021–2022	Arrests Made	Japanese police arrested individuals involved in laundering parts of the stolen NEM.

Source: Compiled from public disclosures, blockchain records, and regulatory reports



XEM Price After The Coincheck Hack

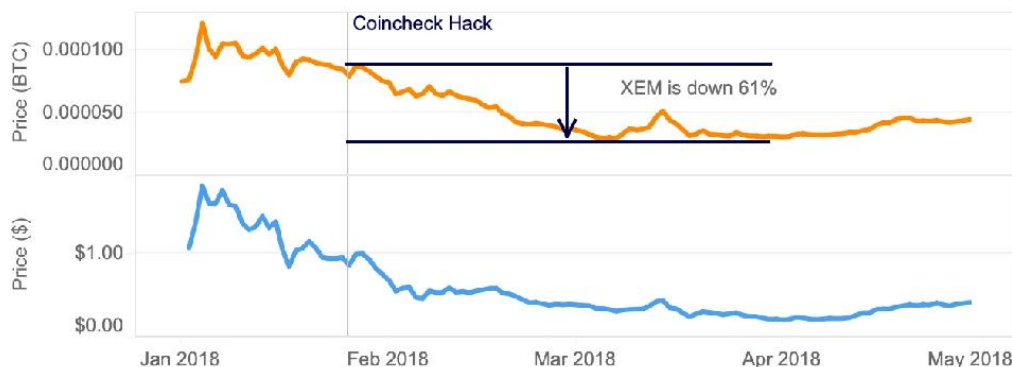


Fig:-3 XEM Price Decline Following the Coincheck Hack (January–May 2018)

ii. Attack Vector Identification and Vulnerability Assessment

Coincheck 2018 Hack — Vulnerability Table

Category	Description of Vulnerability	Impact / How It Helped the Hackers
Hot Wallet Storage	Majority of NEM tokens stored in a hot wallet (online, internet-connected).	Online storage exposed private keys and funds to remote attackers — <i>primary access point for theft.</i>
Single Private Key /	Wallet was protected by a <i>single</i>	Created a <i>single point of failure</i> — once

Category	Description of Vulnerability	Impact / How It Helped the Hackers
No Multi-Sig	private key, lacking multi-signature authorization.	compromised, hackers could transfer all funds.
Poor Internal Security Practices	Weak internal access controls, no segmentation, insufficient employee security training.	Enabled attackers to breach internal systems and obtain hot wallet access.
Malware & Phishing (Probable Initial Entry)	Early phishing emails to employees allegedly installed malware on internal systems.	Gave attackers foothold to harvest credentials/keys <i>weeks before</i> the full theft.
Lack of Real-Time Monitoring & Alerts	No automated alerting for abnormal transactions or large outflows.	Allowed the theft to proceed for hours before detection.
Insufficient Risk & Key Management	No enforced encryption, cold storage protocols, or strict key handling policies.	Exposed private keys to unauthorized access and use.
Regulatory Security Oversight Weakness	Prior lax security compliance and oversight before the hack.	Contributed to systemic weak security practices company-wide.

Despite being a commonly known vulnerability of hot-wallet systems, a closer analysis of the attack that targeted Coincheck has shown several other important attack vector components that were also used to exploit this exchange but were initially overlooked in early media coverage: “there were good indications that the attackers used their initial foothold in Coincheck’s internal network by means of targeted phishing emails that delivered malware to internal workstations, likely with a remote access Trojan that gave the attackers permanent access to workstations of internal staff involved with management of hot-wallets. Having obtained internal access, the attackers were able to use a flat internal network design, where internal workstations, management-level PCs, as well as internal systems with access to private keys are all part of a relatively flat network that enabled easy horizontal movement to higher value systems without necessarily relying on complex exploitation. One of the most profoundly disturbing findings is that the private key for their massive 175,000 Bitcoin NEM hot wallet was being stored on a run-of-the-mill, internet-reachable internal Windows-based desktop computer that was neither encrypted, protected from network access, nor hardwired with

hardware security modules to secure their key usage, which meant that as soon as their internal computer was hacked, either direct access to that key would result in immediate theft, with simultaneous access enabling remote initiation of transactions.” Without multi-user security checks for authorization of key usage as well as lack of enforced internal use separation, this immediately meant that anyone with access to a stolen key effectively received unconditional transaction authorization without human review, which would otherwise serve as a stop-gap for abnormal transactions. Furthermore, in a disturbing indication of a lack of automated security that would otherwise stop a transaction, even when a massive 523,000,000 NEM were transferred out in a remarkably rapid burst of transactions, this still did not result in an emergency lockout of further transactions from this hot-wallet.

V. IMPACT ASSESSMENT

In January 2018, Coincheck, a Tokyo-based exchange, experienced “one of the biggest exchange hacks in history” when hackers stole around 58 billion yen in NEM tokens from their “hot wallet” – leaving a direct Financial impact of hundreds of millions of dollars. Though the

company subsequently announced it would cover most of its users from their own funds (reimbursing tens of billions of yen), this hack, nonetheless, brought it under intense regulatory pressures, an FSA raid, and further functional overhaul that completely changed the face of cryptocurrencies in Japan. On a technical front, Coincheck's vulnerability to this attack has been identified to originate from insecure custody practices, where huge amounts of a certain token, like NEM, were stored in the hot wallet without proper multi-sig security and key management design; public opinion and subsequent investigations also suggested certain social engineering issues that allowed hackers to move funds for many hours before being noticed.



Fig:-4 Market Sell-Off Following Coincheck Trading Suspension
 (January 2018)

The hack's effects, further, went far beyond other "mere" direct Financial losses, with data compromise and risks of downed services being additional concerns that, in total, affected around 260,000 users, "No verified public evidence confirms leakage of customer PII, credentials, or KYC data." halted all other withdrawal and trading of most assets, reputation, and long-term customer distrust that further impacted their volumes to costs of remediation, upgrade, and company overhaul itself, as Coincheck was subsequently acquired by Monex. (No verified public reports confirmed leakage of customer PII, credentials, or KYC data; the incident was limited to unauthorized asset transfers.)

i. Trading Disruption and Financial Exposure

This suspension of trading services caused a significant hindrance to the operations of the users. This is attributed to the fact that the users could not conduct their trading activities or make any adjustments related to their markets in a situation where there was high variability of the markets, which were experiencing high levels of cryptocurrencies price fluctuations. This situation indirectly led to losses of funds even for those whose funds were not actually stolen.

ii. Trust and User Confidence Impact

This event led to a significant loss of trust among users. Being a prominent exchange for cryptocurrencies in Japan, it was expected that Coincheck would implement robust security measures, but it tarnished their reputation to learn that a considerable amount of their assets was stored in a "hot wallet." Uncertainty about the security of their funds as well as the recovery of their accounts escalated the discontent of the users.

iii. User Experience and Regulatory Intervention

This led the FSA to put Coincheck under heightened regulatory scrutiny, compelling the exchange to fix security governance, enhance operational controls, and file periodic compliance reports. These regulatory steps delayed the resumption of full services, resulted in limited product offerings, and involved more stringent KYC/AML requirements for users. While these measures enhanced the long-term security posture of the exchanges, they further prolonged service disruption and adversely impacted short-term user experience.

IV. FORENSIC REPORT

i. Scope and Evidence Limitations

This particular case study's forensic analysis will be based solely on available public data such as the blockchain transaction data, press releases, and statements from the NEM Foundation and Japanese government agencies, and nothing from Coincheck has actually been released regarding the system logs, malware samples, or endpoint forensics. Thus, the following identification of host and malware findings are purely analytic in nature and have been made through open-source intelligence or comparison with other similar breaches at other digital currency exchanges.

ii. Digital Footprint and Blockchain Analysis

The first source of evidence for this incident was the NEM blockchain, which offers immutable record services for all its transactions. Analysis indicated that by utilizing free public blockchain browser tools, NEM could read all its outgoing transactions from its compromised Coincheck hot wallet address, which showed a huge malicious transfer of 523 million XEM coins sent to a malicious address, followed by a rapid shuffling of funds into several secondary wallets.

Graph-based transaction analysis and address clustering methods were utilized to detect behavioral characteristics indicative of money laundering, such as high-velocity money transfer, address fan-out, and consolidation. The NEM Foundation's address tagging project and exchange alert notifications from OSINT helped to achieve partial attribution to money laundering routes and attempted launder-to-altcoin exchanges.

iii. OSINT-Based Attribution and Activity Correlation

The open-source intelligence analysis techniques that enriched the blockchain information were used to link public warnings by the NEM Foundation, exchange blacklists, public announcements of regulations, and reports of suspect activity on the dark net to activity on the blockchain to pinpoint possible cash-out channels and middlemen services. Although it was not possible to directly attribute this attack to its perpetrators, it helped to point out the possible money transfer routes that could then be investigated by law enforcement.

iv. Malware and Initial Access Analysis (Inferred)

Public investigations and industry analyses suggest that the attackers may have gained an initial foothold through targeted phishing emails sent to Coincheck employees, potentially resulting in the deployment of a Remote Access Trojan (RAT). This malware is believed to have enabled credential harvesting, lateral movement, and unauthorized access to wallet management systems. However, no confirmed malware samples or indicators of compromise were publicly released by Coincheck, and this assessment is based on inference from attack characteristics, timing, and similarities to previously documented exchange breaches.

V. Forensic Tools

Forensic Area	Tool / Method	Purpose
Blockchain Analysis	NEM Blockchain Explorer	Trace stolen fund movements
Transaction Visualization	Graph analysis techniques	Identify laundering patterns
OSINT Collection	OSINT Framework, public disclosures	Contextual enrichment

Log Correlation	Timeline reconstruction	Align internal events with on-chain activity
Attribution Support	Exchange alerts, tagging systems	Identify conversion attempts

vi. Log and Transaction

While the logs obtained through internal system activity remained unseen, an internal system log correlation timeline could be asserted through the matching of timestamp information on the blockchain with publicly known operational event actions, including the cessation of trading activities and frozen withdrawals. The internal system access and blockchain financial transaction timing match makes the argument that it is not the blockchain protocol that resulted in unauthorized financial transactions, through the means of credentials or internal tools

V. Technical Mitigation & Security Recommendations

i. Immediate Actions

First, as soon as Coincheck noticed the unauthorized transfer of 523 million XEM, they should have implemented an immediate response plan and therefore immediately set all outgoing transactions from the hot wallets to unavailable, removed the compromised servers from the network, and changed all potentially compromised passwords. There should be an immediate chain-of-custody incident response, involving an efficient preservation and subsequent backups of all system logs, wallet login logs, and endpoint memory images. Moreover, handling public communications should have followed an efficient incident response script, which should start with an honest and open early breach notification, an immediate suspend operation on all deposits and

withdrawals on all assets, cooperation with the NEM Foundation tagging system, and simultaneous updates on all customer, government, and law enforcement updates.

i. Long-Term Security Measures

To prevent similar large-scale losses, Coincheck should have transitioned from single-signature hot wallets to a **multi-signature, hardware-secured wallet architecture** with strict transaction-approval workflows. High-value assets should have been moved to **cold storage**, with hot wallets maintaining only the minimum operational liquidity. The exchange needed a modern **Zero-Trust security model** with continuous authentication, least-privilege access, MFA for all internal systems, and periodic credential rotation. Infrastructure redesign must include **network segmentation**, dedicated keys stored in HSMs, and tamper-resistant audit controls. Comprehensive **security policies** should mandate regular red-team assessments, mandatory phishing-resistance training for employees, vulnerability scanning, and third-party security audits. Advanced **monitoring and detection systems**—such as SIEM with real-time anomaly alerts, on-chain transaction monitoring, automated withdrawal-velocity detection, and blockchain-based risk scoring—would drastically reduce response time. Finally, a formal **incident-response playbook**, coordinated with Japan's FSA and the global crypto-compliance ecosystem, ensures operational resilience and immediate activation of countermeasures in future breaches.

Executive-Level Mitigation Plan

This executive-level mitigation plan outlines phased corrective actions designed to reduce the likelihood, impact, and recurrence of large-scale cryptocurrency exchange breaches. The focus is on asset custody security, access control, monitoring, governance, and regulatory alignment, presented in a manner suitable for

senior management and non-technical stakeholders.

1. Immediate Mitigation Actions (0–30 Days)

Objective: Contain damage, stabilize operations, and prevent further unauthorized activity following breach detection.

Control Area	Executive Action	Risk Addressed
Asset Containment	Immediately suspend all non-essential hot-wallet withdrawals	Prevents additional unauthorized fund movement
Credential & Key Security	Rotate all private keys and revoke potentially compromised credentials	Eliminates attacker persistence
Incident Governance	Activate executive incident response leadership and regulator communication channels	Reduces regulatory and reputational exposure
Evidence Preservation	Secure and preserve system logs, wallet metadata, and transaction records	Enables forensic and legal investigation
Customer Communication	Issue transparent breach notifications and define refund mechanisms	Maintains customer confidence

Executive Insight: Rapid containment and transparent communication significantly reduce financial and reputational damage.

2. Short-Term Mitigation Measures (1–6 Months)

Objective: Eliminate structural weaknesses that enabled the breach.

Control Area	Mitigation Measure	Security Benefit
Wallet Architecture	Migrate at least 95% of digital assets to cold storage	Minimizes exposure to online attacks
Transaction Authorization	Enforce multi-signature approval for all high-value transactions	Removes single-point-of-failure
Key Management	Deploy Hardware Security Modules (HSMs) for private key storage	Prevents key exfiltration
Monitoring & Alerts	Implement real-time blockchain and withdrawal anomaly detection	Enables early breach detection
Internal Access Controls	Enforce MFA, least privilege access, and role separation	Limits insider and phishing risks
Security Awareness	Conduct mandatory phishing and security awareness training	Reduces social engineering success

Executive Insight: Most exchange breaches occur due to weak custody architecture rather than blockchain protocol flaws.

3. Long-Term Strategic Controls (6–12 Months)

Objective: Establish a resilient, regulator-aligned security posture.

Strategic Area	Executive Initiative	Business Impact
Zero-Trust Security	Implement continuous authentication and network segmentation	Prevents lateral movement
Governance & Oversight	Establish board-level security risk reporting	Improves accountability
Regulatory Alignment	Align controls with ISO 27001, NIST CSF, and FSA requirements	Reduces compliance risk
Continuous Testing	Perform periodic red-team and wallet compromise simulations	Validates control effectiveness
Incident Readiness	Maintain tested incident response playbooks	Reduces recovery time
Independent Audits	Conduct third-party security and custody audits	Enhances stakeholder trust

Executive Insight: Security maturity directly impacts exchange licensing, valuation, and long-term survival.

4. Risk Reduction Summary

Risk Category	Pre-Mitigation Risk	Post-Mitigation Risk
Hot wallet compromise	High	Low
Private key theft	High	Low
Undetected fund drainage	High	Medium–Low
Regulatory penalties	High	Low
Customer trust erosion	High	Medium

5. Executive Key Takeaways

- Cryptocurrency exchange breaches are primarily custody and governance failures.
- Hot wallets must be treated as high-risk operational systems.
- Early detection and response significantly reduce financial losses.
- Regulatory engagement is as critical as technical remediation.
- Strong security governance is a business-critical requirement.

VI. Forensic Evidence Portfolio

This section documents all evidentiary artifacts used to support the forensic analysis of the Coincheck 2018 cryptocurrency exchange breach. All evidence presented is derived from **publicly verifiable sources**, including blockchain explorers, regulatory disclosures, and reputable media reports. No internal Coincheck system logs or proprietary forensic artifacts were publicly released; therefore, conclusions are

based on **open-source intelligence (OSINT)** and **blockchain transaction analysis**.

Tool-to-Evidence Mapping

Public NEM blockchain explorers were used to confirm unauthorized transactions, attacker wallets, and fund dispersion patterns (A1–A3). OSINT sources validated the incident timeline, disclosure, and regulatory response (A4–A6). Host-based forensic tools were not applicable due to the absence of publicly released internal artifacts.

Evidence ID	Evidence Type	Source	Date (UTC/JST)	Description	Forensic Relevance / What It Confirms
A1	Blockchain Transaction Screenshot	NEM Blockchain Explorer	26 Jan 2018 (JST)	Screenshot of an unauthorized outgoing transaction transferring approximately 523 million XEM from a Coincheck-controlled hot wallet to an external address.	Confirms unauthorized asset transfer , exact loss magnitude , transaction timestamp, and originating wallet linked to Coincheck custody.
A2	Wallet Address Overview	NEM Blockchain Explorer	26 Jan 2018 (JST)	Overview page of the attacker-controlled wallet showing rapid accumulation of stolen XEM immediately after the breach.	Demonstrates attacker-controlled destination address and abnormal balance increase consistent with a theft event.
A3	Transaction Graph / Fund Flow Visualization	NEM Blockchain Explorer (Graph View)	26–28 Jan 2018	Visualization showing rapid fan-out and fragmentation of stolen XEM across multiple secondary wallets.	Provides evidence of money laundering behavior , including fund obfuscation and anti-tracing techniques.
A4	Public Incident Disclosure (Media Report)	Reuters	26 Jan 2018	News article confirming Coincheck’s public admission of the hack and estimated losses of approximately US\$530 million .	Corroborates timeline of detection and disclosure , validating forensic timing derived from blockchain data.
A5	Regulatory Enforcement Notice	Financial Services Agency (Japan)	Feb–Mar 2018	Official FSA business improvement order issued to Coincheck following the breach.	Confirms governance failures , regulatory findings, and institutional response to security control weaknesses.

A6	Blockchain Address Tagging Announcement	NEM Foundation	Jan–Feb 2018	Public statement describing on-chain tagging of attacker-controlled addresses and exchange coordination efforts.	Supports community-led forensic response , attempted fund containment, and collaborative mitigation actions.
A7	Market Impact Visualization (Optional)	Public Market Data Sources	Jan–Feb 2018	Price and volume changes in XEM following trading suspension and breach disclosure.	Demonstrates secondary financial and market impact , supporting the broader impact assessment.
A8	Law Enforcement Disclosure	Metropolitan Police Department (Japan)	Jan 2021	Announcement of arrests related to laundering stolen NEM while knowing the assets were illicit.	Confirms partial attribution , downstream laundering activity, and long-term investigative outcomes.

Evidence Integrity & Limitations Statement

All evidence presented above is sourced from **publicly accessible and verifiable materials**. No internal Coincheck system logs, endpoint images, malware samples, or private key artifacts were released publicly. Consequently, **host-level compromise and malware analysis findings are inferred**, based on attack patterns, timing correlations, and similarities to documented cryptocurrency exchange breaches.

Blockchain evidence is considered **high-integrity**, as transaction records are immutable and timestamped; however, **blockchain data alone cannot attribute attacker identity**, only activity.

REFERENCES

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[6] Nikkei Asia, “*Coincheck Hack Exposes Weak Security Practices at Crypto Exchanges*,” 2018.

[7] Chainalysis, “Cryptocurrency Crime and Exchange Hacks: Lessons from Coincheck,” Industry Analysis Report, 2019.

[8] Japan Times, “Coincheck to Reimburse Users after Record Cryptocurrency Theft,” January 2018.

APPENDIX

The following appendix contains publicly available forensic information used in this analysis for Coincheck Cryptocurrency Exchange hacking in 2018. Such information is gathered using blockchain transaction information and OSINT information. Coincheck systems were not compromised, and any non-public information did not come from their systems. The information in this appendix confirms information presented in this analysis with regards to timeline, financial damage, attack behavior, and reactions

Figure A1. Unauthorized NEM Transfer from Coincheck Hot Wallet

Figure A1. illustrates a public blockchain transaction record from the NEM blockchain network, which verifies an unauthorized outgoing transfer of about 523 million XEM from a Coincheck-operated hot wallet on January 26th, 2018. The time stamp, amount of money transferred, and origin and destination wallets all support the magnitude of this financial loss.

Source: NEM Blockchain Explorer

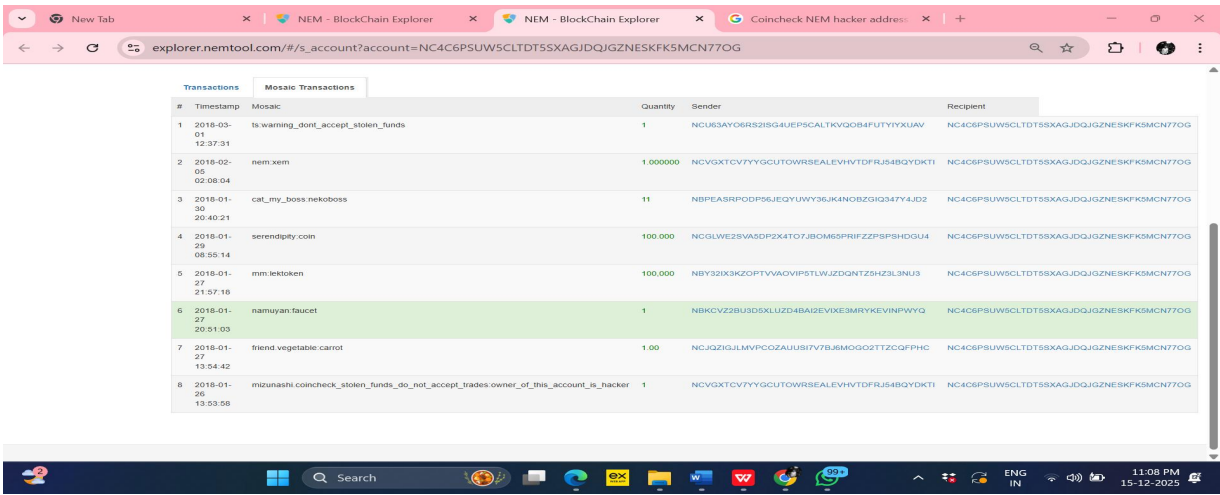


Figure A2. Attacker-Controlled Wallet Address Receiving Stolen NEM

Figure A2: Blockchain address to which the stolen NEM tokens were initially transferred. Address overview showing strange balance building up and a lot of inbound transfers in a very short period of time after the breach, which are indicative of an attacker in control.

Source: NEM Blockchain Explore

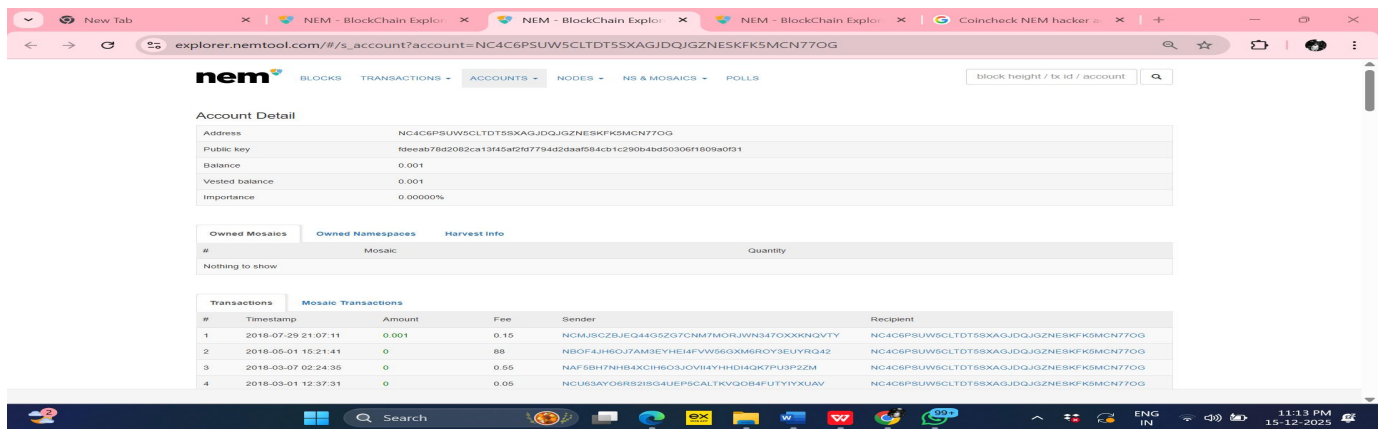


Figure A3. Fragmentation of Stolen NEM Tokens Across Multiple Addresses

Figure A3. shows how quickly the stolen NEM tokens have been distributed from the main attacker’s wallet into other secondary wallets. Furthermore, this pattern of fragmentation of wallets is common in money laundering schemes used to make transactions not easily traceable

Source: NEM Blockchain Explorer / Transaction Graph View

“Following the breach, the NEM Foundation rapidly deployed on-chain tagging mechanisms to label attacker-associated addresses with warning mosaics, effectively restricting the circulation of stolen funds (see Figure A6).”

Figure A4. Public Disclosure and Incident Confirmation

Figure A4 shows a news article publicly announcing the hack on Coincheck, including the date when it was detected and the estimated losses of approximately 530 million US dollars. With regards to blockchain time stamps, this serves to prove the time lag observed before making the information publicly available.

Source: Reuters, January 2018

Figure A5. Regulatory Enforcement Action by Japan’s Financial Services Agency

Figure A5 illustrates an official notice issued by the Financial Services Agency of Japan, FSA, instructing Coincheck to make improvements in business and security measures after the occurrence. The above piece of evidence confirms findings concerning governance issues and risk management controls.

Source: Financial Services Agency (Japan)

Figure A6. NEM Foundation Address Tagging and Exchange Coordination (Optional)

Figure A6 shows a statement from the NEM Foundation announcing the marking of attacker-controlled addresses and cooperation with exchanges to limit the mobility of illicit funds. Such an effort shows case analysis facilitated by blockchain forensics conducted by the community.

Source: NEM Foundation Public Announcement.

explorer.nemtool.com/#/s_account?account=NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG

Transactions

Mosaic Transactions

#	Timestamp	Mosaic	Quantity	Sender	Recipient
1	2018-03-01 12:37:31	ts.warning_dont_accept_stolen_funds	1	NCU63AYO6RS2ISG4UEP5CALTkvQOB4FUTYIYXUAV	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
2	2018-02-05 02:08:04	nem:xem	1.000000	NCVGXTCV7YYGCUTOWRSEALEVHVTDfRJ54BQYDKtI	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
3	2018-01-30 20:40:21	cat_my_boss.nekoboss	11	NBPEASRPDPF56JEQYUWY36JK4NOBZGIQ347Y4JD2	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
4	2018-01-29 08:55:14	serendipity.coin	100.000	NCGLWE2SVA5DP2X4T07JBOM65PRIFZZPSPSHDGu4	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
5	2018-01-27 21:57:18	mm.lektoken	100.000	NBY32IX3KZOPTVVAOVIP5TLWJZDQNTZ5HZ3L3NU3	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
6	2018-01-27 20:51:03	namuyan.faucet	1	NBKCVZ2BU3D5XLUZD4BAI2EVIXE3MRyKEVINPwYQ	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
7	2018-01-27 13:54:42	friend.vegetable.carrot	1.00	NCJQZIGJLMVPCOZAUUSI7V7BJ6MOGO2TTZCQFPHC	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG
8	2018-01-26 13:53:58	mizunashi.coincheck_stolen_funds_do_not_accept_trades.owner_of_this_account_is_hacker	1	NCVGXTCV7YYGCUTOWRSEALEVHVTDfRJ54BQYDKtI	NC4C6PSUW5CLTDT5SXAGJDQJGZNESKFK5MCN77OG

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